

Math 166

Names: _____

New series from old

Section Number: _____

For these exercises, you may use the table of common Maclaurin series from the notes and from Table 1 in Section 11.10 of your textbook.

1. What is $\pi - \frac{\pi^3}{3!} + \frac{\pi^5}{5!} - \frac{\pi^7}{7!} + \cdots$?

2. Which function has Maclaurin series $\sum_{n=0}^{\infty} (-1)^n 5^n x^n$?

3. Recall that $\sin(x^2)$ has no antiderivative (in terms of elementary functions). In this problem you will approximate $\int_0^1 \sin(x^2) dx$ using Taylor series.
- (a) Write the Maclaurin series of $\sin(x)$ then substitute x^2 for x to obtain a Maclaurin series for $\sin(x^2)$.
 - (b) Integrate this series to find the Maclaurin series of $\int \sin(x^2) dx$. Remember the $+C$ constant after integrating.
 - (c) Use the above to find the 7th order Taylor polynomial for $\int \sin(x^2) dx$.
 - (d) Use this Taylor polynomial to find a numerical approximation for the definite integral $\int_0^1 \sin(x^2) dx$. What happens to the constant from part (b)?
 - (e) Use relevant error bounds to find the smallest value of n for which the Taylor polynomial of order n for $\int \sin(x^2) dx$ is guaranteed to have an error less than 0.1 when used to approximate $\int_0^1 \sin(x^2) dx$.